

Missing Data and Imputation

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Introduction

This markdown tackles ad-hoc imputation methods as well as multiple imputation, all using the `mice` package.

The following steps are performed:

- Data processing
- Inspection of the missingness pattern
- Apply ad hoc imputation methods using the `mice` package
- Apply multiple imputation using the `mice` package
- Perform linear regression and compare the output when different imputation approaches are used

Load the required libraries.

```
# Load libraries
library(tidyverse)
library(mice)
library(ggmice)
library(ISLR)

# set seed for reproducability
set.seed(123)
```

```
# set scippen to suppress scientific notations
options(scipen = 999)
```

Data Processing

From the ISLR package, We will use the College dataset and the following variables:

- Out-of-state tuition
- PhD: Pct. of faculty with Ph.D.'s
- Terminal: Pct. of faculty with terminal degree
- Expend: Instructional expenditure per student

```
data_complete <-
  College |>
  select(Outstate, PhD, Terminal, Expend)

summary(data_complete)
```

##	Outstate	PhD	Terminal	Expend
## Min.	: 2340	Min. : 8.00	Min. : 24.0	Min. : 3186
## 1st Qu.:	7320	1st Qu.: 62.00	1st Qu.: 71.0	1st Qu.: 6751
## Median :	9990	Median : 75.00	Median : 82.0	Median : 8377
## Mean :	10441	Mean : 72.66	Mean : 79.7	Mean : 9660
## 3rd Qu.:	12925	3rd Qu.: 85.00	3rd Qu.: 92.0	3rd Qu.:10830
## Max.	:21700	Max. :103.00	Max. :100.0	Max. :56233

We focus on how different imputation approaches affect the linear relationship between `Outstate` and `Expend`. First we inspect how the relationship looks like on the complete dataset.

```
lm_complete <- lm(Expend ~ Outstate,
  data = data_complete)

summary(lm_complete)
```

```
##
## Call:
## lm(formula = Expend ~ Outstate, data = data_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6131  -2022   -637   1027   39273
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 542.86969  385.92915   1.407        0.16
## Outstate      0.87325    0.03449  25.315 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3866 on 775 degrees of freedom
## Multiple R-squared:  0.4526, Adjusted R-squared:  0.4519
## F-statistic: 640.9 on 1 and 775 DF, p-value: < 0.00000000000000022
```

We can now generate missingness in the variable `Expend` and `Terminal` under a Seen Data Dependent (SDD) mechanism, assuming that the missingness in these variables are dependent on the values of the `Outstate` and `PhD` respectively

```
# logistic function
logistic <- function(x) (exp(x) / (1 + exp(x)))

# number of observations
N <- nrow(data_complete)

# predictor Outstate
misprob_out <- logistic(scale(data_complete$Outstate))

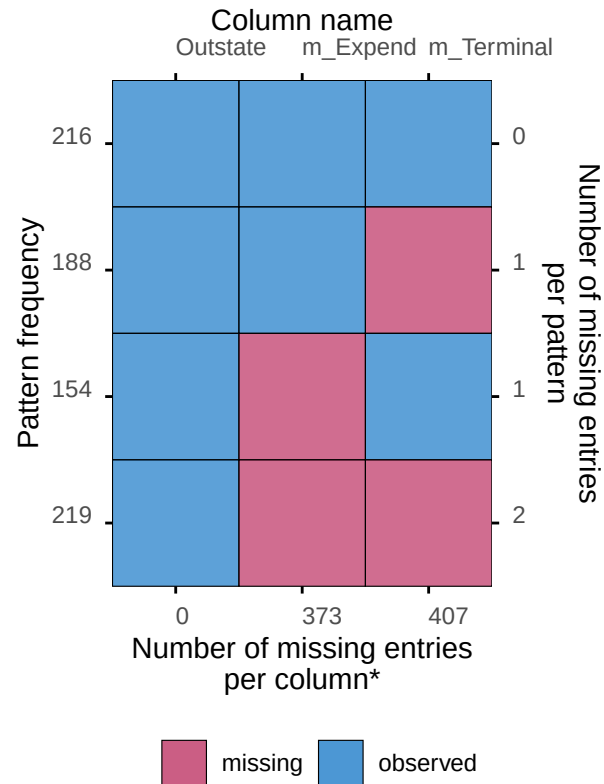
# predictor PhD
misprob_phd <- logistic(scale(data_complete$PhD))

data_missing <-
  data_complete |>
  mutate(
    # response indicator outstate
    R_out = 1 - rbinom(N, 1, misprob_out),
    # response indicator PhD
    R_phd = 1 - rbinom(N, 1, misprob_phd),
    # Missing in expend
    m_Expend = ifelse(R_out == 0, NA, Expend),
    # and in terminal
    m_Terminal = ifelse(R_phd == 0, NA, Terminal)
  ) |>
  # Select variables that we will use
  select(Outstate, m_Expend, m_Terminal)
```

Inspection of missingness

Inspect the missingness pattern using `ggmice` package function

```
plot_pattern(data_missing)
```



*total number of missing entries: 780

Ad Hoc Imputation Methods

Listwise Deletion

```
lm_listwise <- lm(m_Expend ~ Outstate,
                  data = data_missing)
```

```
summary(lm_listwise)
```

```
##
## Call:
## lm(formula = m_Expend ~ Outstate, data = data_missing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4986.2 -1490.9  -398.1   736.4 30053.4
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 2006.74472   396.63965    5.059 0.000000641 ***
## Outstate      0.70211     0.04153   16.904 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 2758 on 402 degrees of freedom
## (373 observations deleted due to missingness)
## Multiple R-squared: 0.4155, Adjusted R-squared: 0.414
## F-statistic: 285.8 on 1 and 402 DF, p-value: < 0.00000000000000022
```

*# The coefficient is a bit smaller, the standard error somewhat larger, so we
introduced some bias, and reduced the power. Both could have been expected,
due to how we generated missings and since we lost 373 cases.*

Regression Imputation

```
imp_regression <- mice(
  data = data_missing,
  method = "norm.predict",
  m = 1,
  maxit = 1
)
```

```
##
## iter imp variable
## 1 1 m_Expend m_Terminal
```

`complete()` takes an object, fills in the missing data, and returns the complete data in a specified format.

```
complete(imp_regression) |>
  is.na() |>
  sum()
```

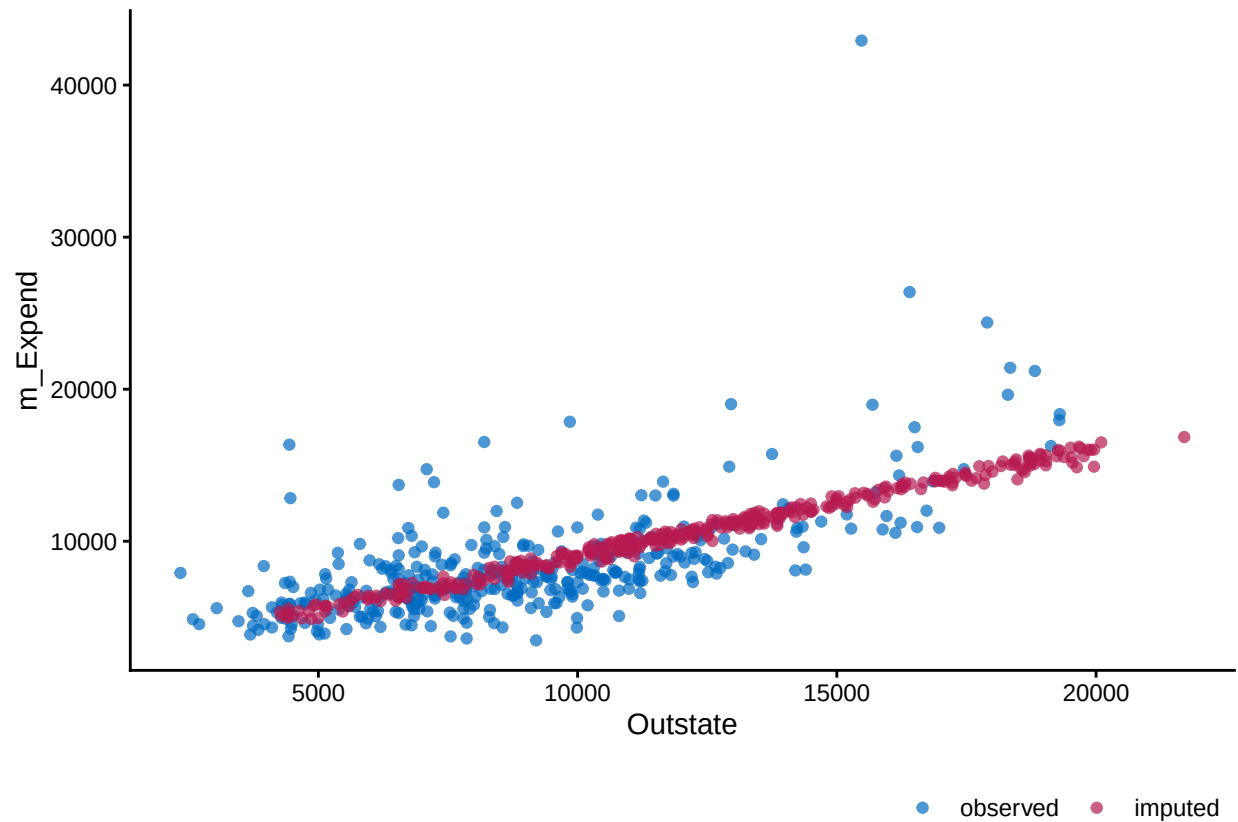
```
## [1] 0
```

```
lm_regression <- with(imp_regression, lm(m_Expend ~ Outstate))

summary(lm_regression)
```

```
## # A tibble: 2 x 7
##   term          estimate std.error statistic   p.value  nobs df.residual
##   <chr>          <dbl>    <dbl>    <dbl>   <dbl> <int>    <dbl>
## 1 (Intercept) 1989.      199.      9.97 4.05e- 22   777      775
## 2 Outstate    0.703     0.0178   39.5 1.44e-187   777      775
```

```
imp_regression |>
  ggmlm(aes(x = Outstate, y = m_Expend)) +
  geom_point()
```



```
# Almost all imputed values are on a (almost) straight regression line (which
# is how we specified the imputation model). The variation around this line
# comes from the other variables that are also used in calculating the
# predictions.
```

Stochastic Regression Imputation

```
imp_stochastic <- mice(
  data   = data_missing,
  method = "norm.nob",
  m      = 1,
  maxit  = 1
)

##
## iter imp variable
##   1   1 m_Expend m_Terminal

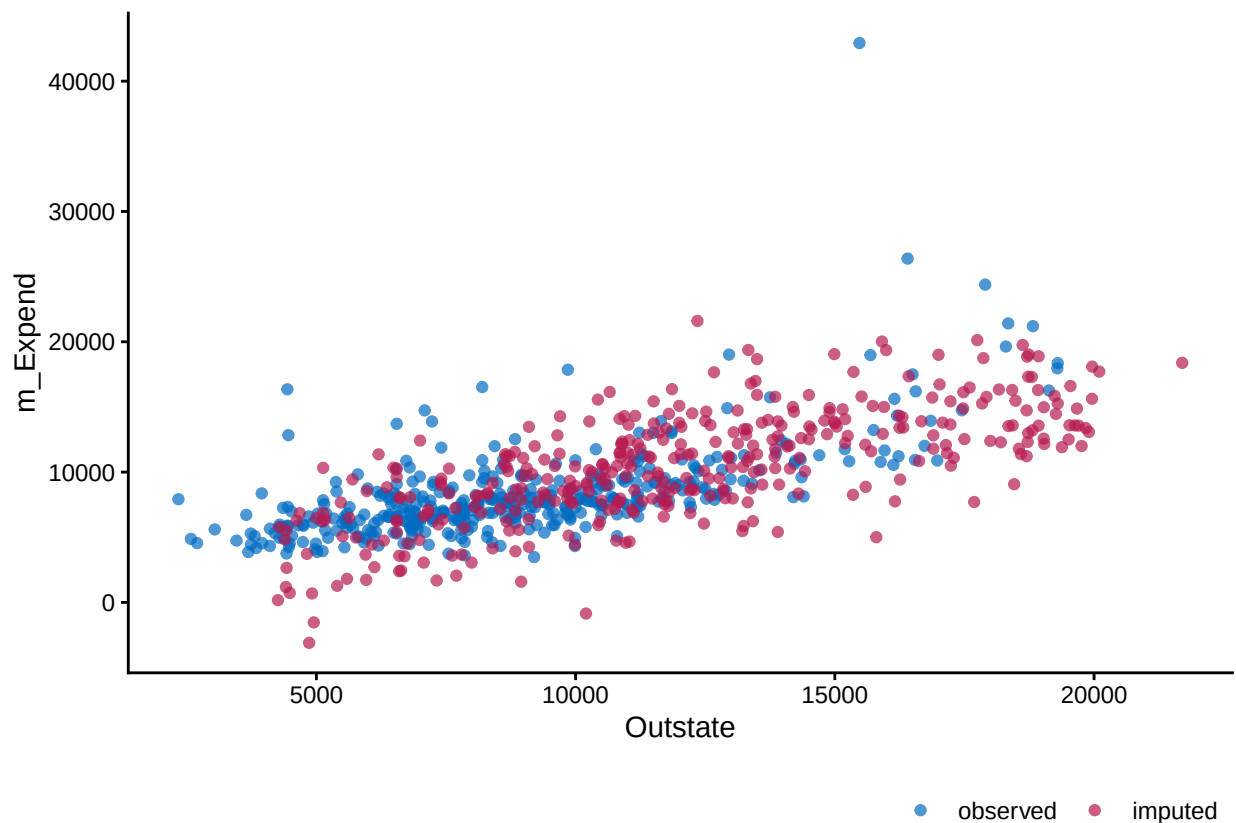
lm_stochastic <- with(imp_stochastic, lm(m_Expend ~ Outstate))

summary(lm_stochastic)

## # A tibble: 2 x 7
```

```
##      term      estimate std.error statistic    p.value  nobs df.residual
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl> <int>    <dbl>
## 1 (Intercept) 1873.      287.      6.53 1.22e- 10   777      775
## 2 Outstate      0.715    0.0257    27.9 6.66e-119   777      775
```

```
imp_stochastic |>
  ggmlize(aes(x = Outstate, y = m_Expend)) +
  geom_point()
```



There is considerably more variation between the imputed values as compared to the regression imputations. However, the current approach still treats the estimated regression coefficients as fixed, although there is also uncertainty with respect to these coefficients. This uncertainty is still not taken into account with stochastic regression imputation.

Multiple Imputation

```
imp_multiple <- mice(data_missing)
```

```
##
## iter imp variable
##   1   1 m_Expend m_Terminal
##   1   2 m_Expend m_Terminal
```

```
## 1 3 m_Expend m_Terminal
## 1 4 m_Expend m_Terminal
## 1 5 m_Expend m_Terminal
## 2 1 m_Expend m_Terminal
## 2 2 m_Expend m_Terminal
## 2 3 m_Expend m_Terminal
## 2 4 m_Expend m_Terminal
## 2 5 m_Expend m_Terminal
## 3 1 m_Expend m_Terminal
## 3 2 m_Expend m_Terminal
## 3 3 m_Expend m_Terminal
## 3 4 m_Expend m_Terminal
## 3 5 m_Expend m_Terminal
## 4 1 m_Expend m_Terminal
## 4 2 m_Expend m_Terminal
## 4 3 m_Expend m_Terminal
## 4 4 m_Expend m_Terminal
## 4 5 m_Expend m_Terminal
## 5 1 m_Expend m_Terminal
## 5 2 m_Expend m_Terminal
## 5 3 m_Expend m_Terminal
## 5 4 m_Expend m_Terminal
## 5 5 m_Expend m_Terminal
```

```
imp_multiple
```

```
## Class: mids
## Number of multiple imputations: 5
## Imputation methods:
##   Outstate  m_Expend m_Terminal
##         ""      "pmm"      "pmm"
## PredictorMatrix:
##           Outstate m_Expend m_Terminal
## Outstate         0         1         1
## m_Expend         1         0         1
## m_Terminal        1         1         0
```

```
lm_multiple <- with(imp_multiple, lm(m_Expend ~ Outstate))
```

```
# The `lm_multiple` object contains the regression summaries for each of the 5
# datasets. The new object `lm_multiple` is actually of class `mira`, which
# stands for Multiply Imputed Repeated Analyses.
# in the real data.
```

```
pool_multiple <- pool(lm_multiple)
```

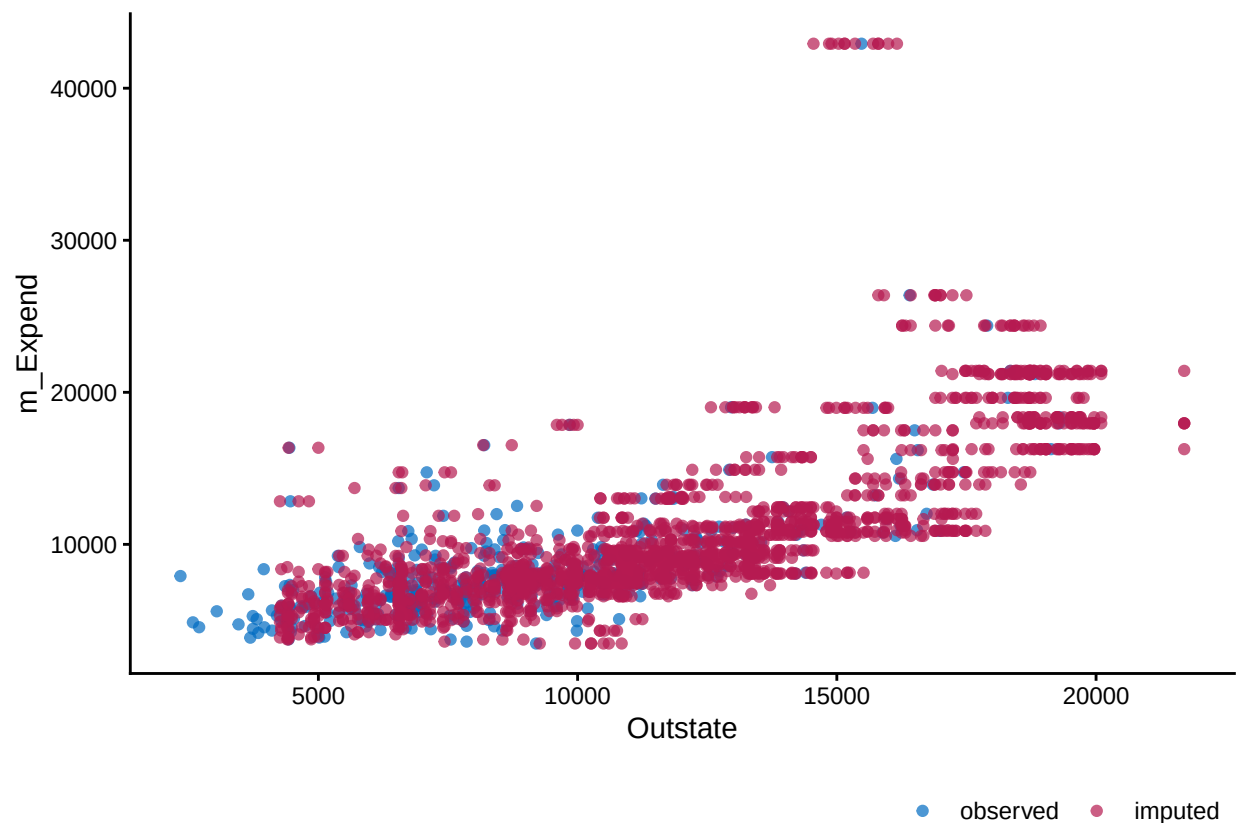
```
summary(pool_multiple)
```

```
##           term      estimate  std.error statistic      df      p.value
## 1 (Intercept) 660.1969423 465.57223416  1.418033 14.80134 0.1768906399981168
## 2   Outstate   0.8532637   0.04434059 19.243399 11.95165 0.0000000002327278
```

```
# These results show the regression coefficient and corresponding standard
# error obtained after pooling the estimates from the five regression models
# applied on the five imputed datasets.

# The complete pooled object gives the relevant pooled regression coefficients
# and parameters, as well as the fraction of information about the coefficients
# missing due to nonresponse (`fmi`) and the proportion of the variation
# attributable to the missing data (`lambda`). The pooled fit object is of
# class `mipo`, which stands for multiply imputed pooled object.*
```

```
imp_multiple |>
  ggmlc(aes(x = Outstate, y = m_Expend)) +
  geom_point()
```



```
# Here we take all uncertainty with respect to the imputations into account
# (i.e., uncertainty with respect to the values and with respect to the
# coefficients). Moreover, we impute multiple times, so that we can assess the
# variability between different data sets (on the level of the estimated model
# parameters). This is often the best approach when interest is in making
# inferences with respect to the imputed data.
```

Comparisson

```
data.frame("method" = c("complete data",
                        "listwise deletion",
                        "regression imputation",
                        "stochastic regression imputation",
                        "multiple imputation"),
          "coefficient" = c(summary(lm_complete)$coefficients[2,1],
                           summary(lm_listwise)$coefficients[2,1],
                           summary(lm_regression)$estimate[2],
                           summary(lm_stochastic)$estimate[2],
                           summary(pool_multiple)$estimate[2]),
          "standard error" = c(summary(lm_complete)$coefficients[2,2],
                               summary(lm_listwise)$coefficients[2,2],
                               summary(lm_regression)$std.error[2],
                               summary(lm_stochastic)$std.error[2],
                               summary(pool_multiple)$std.error[2]))
```

```
##                method coefficient standard.error
## 1             complete data   0.8732488      0.03449491
## 2             listwise deletion   0.7021111      0.04153479
## 3             regression imputation   0.7034919      0.01782889
## 4 stochastic regression imputation   0.7148659      0.02565823
## 5             multiple imputation   0.8532637      0.04434059
```

*# When comparing the results from the different approaches, it can be
concluded that in case of a dataset where the missingness follows the
mechanism Seen Data Dependent, the estimate obtained after applying multiple
imputation is closest to the original estimate obtained from the complete
dataset. In addition, it can be concluded that the standard error obtained
using multiple imputation is somewhat larger than the standard error obtain
from the complete dataset. This is due to the fact that when the standard
errors are pooled over the different imputations, not only the average is
taken, but the variance over the differences is also incorporated (check
the pooling rules for more detail), i.e. the standard error also incorporates
uncertainty due to the missing values. In contrast, the results obtained
using listwise deletion, regression imputation and stochastic imputation all
show a biased estimate, while regression imputation and stochastic
imputation also show a standard error that is too low.**

^